

BIOS vs UEFI Comparison

ITEP 414 – System Administration and Maintenance
Week 3 Portfolio Project
Enterprise Server Deployment and Operating System Installation

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1. Comparison Table

Category	BIOS	UEFI
Definition	Legacy firmware used to initialize hardware and start the OS.	Modern standardized firmware interface for initialization and boot applications.
Boot Process	POST → boot device → MBR boot code → boot loader.	Firmware initialization → EFI boot application/boot loader → OS.
Maximum Disk Support	Traditionally limited by MBR to about 2 TB.	Works with GPT and supports much larger disks.
Partition Style	MBR	GPT
Security	Limited compared with modern UEFI.	Supports Secure Boot and other modern security capabilities.
Speed	Less flexible on modern hardware.	Designed for efficient, flexible modern booting.
Advantages	Simple and compatible with legacy systems.	Large-disk support, GPT, Secure Boot, more partitions, extensibility.
Disadvantages	Legacy storage and security limitations.	More complex; older systems may not support it.
Modern Usage	Primarily legacy systems.	Standard for most modern computers.

2. Why UEFI Has Largely Replaced BIOS

BIOS is the traditional firmware technology used to initialize hardware and start an operating system. It commonly relies on the Master Boot Record (MBR), which has limitations such as the traditional approximately 2 TB disk-size limit and fewer partitioning capabilities.

UEFI provides a more modern and extensible pre-boot environment. It works well with the GUID Partition Table (GPT), which supports much larger storage devices and many more partitions. UEFI also supports technologies such as Secure Boot, helping protect the boot process from unauthorized boot software.

Modern computers require larger storage, stronger security, and more flexible boot management. For these reasons, UEFI has largely replaced legacy BIOS. BIOS knowledge remains useful when administering older computers and legacy systems.

3. Key Takeaway

BIOS remains important for understanding and administering legacy systems, while UEFI is the modern standard because it provides GPT support, larger-disk capability, stronger boot security features, and a more flexible boot environment.

Reference: UEFI Forum — <https://uefi.org/>